SSH on Windows

Step 1: Open Git Bash

On Windows:

- Click **Start**
- Search **Git Bash**
- Open it (not CMD, not PowerShell)

Step 2: Generate SSH key

In **Git Bash**, run:

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

Use the same email as your GitHub account.

When it asks:

```
Enter file in which to save the key:
```

Just press **Enter**

```
Enter passphrase:
```

Press **Enter** (or set one if you want extra security)

SSH key created.

Step 3: Start SSH agent & add key

Run these **one by one**:

```
eval "$(ssh-agent -s)"
```

```
ssh-add ~/.ssh/id_ed25519
```

"Identity added"

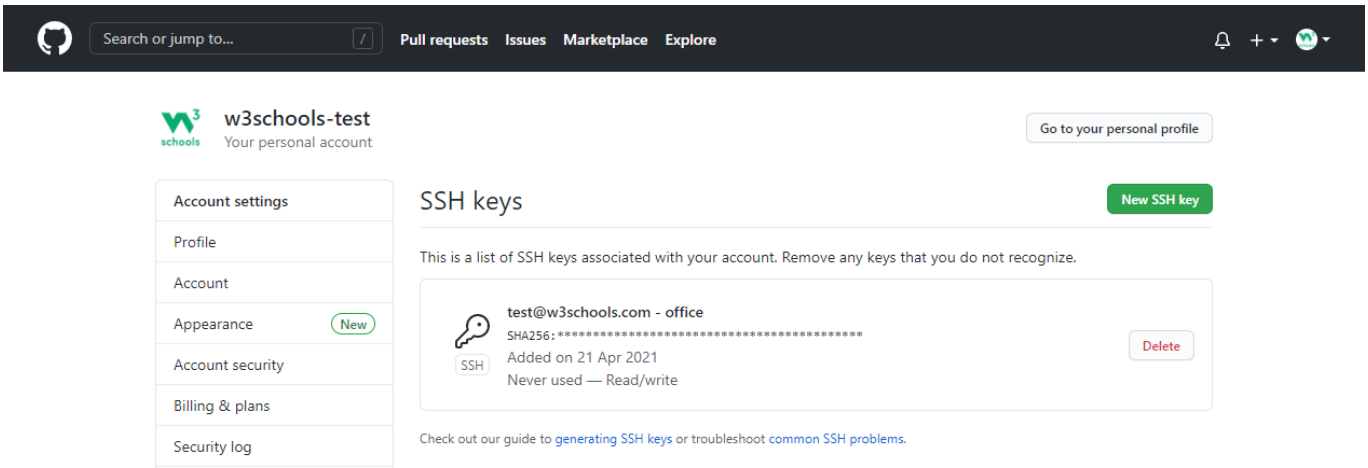
Step 4: Copy SSH public key

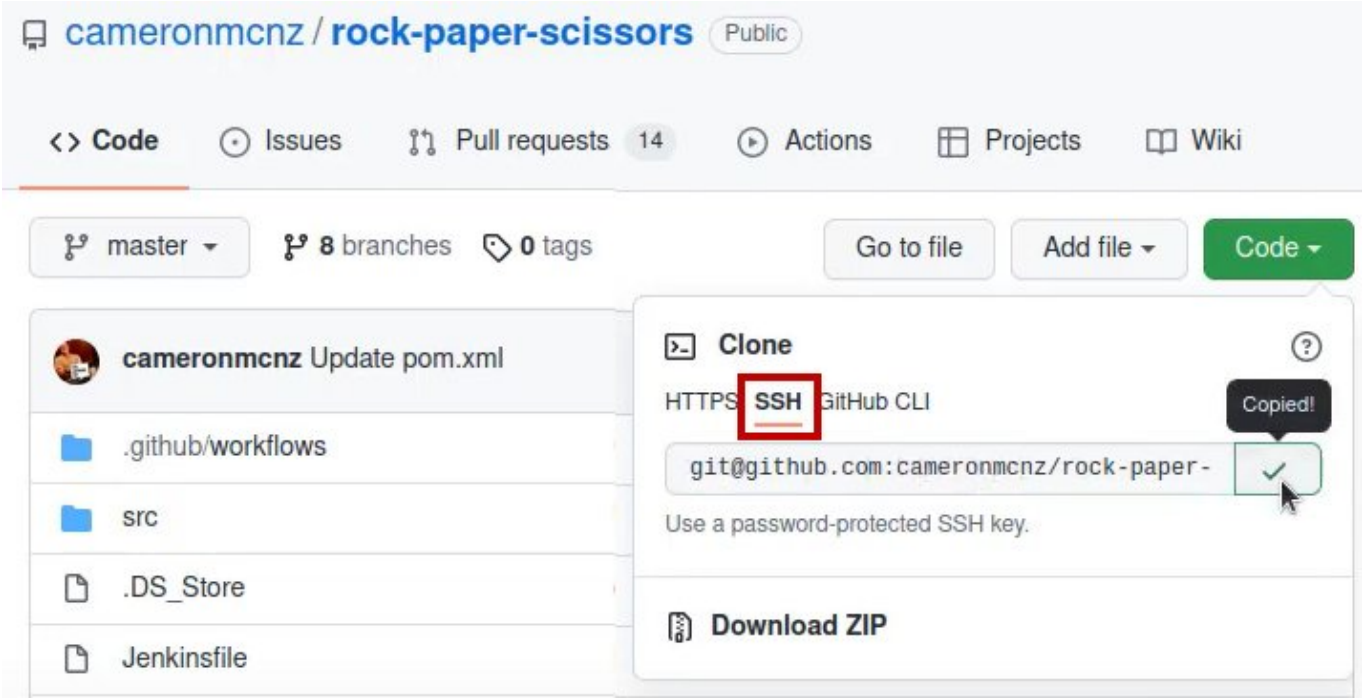
```
cat ~/.ssh/id_ed25519.pub
```

- Copy the **entire output** (starts with `ssh-ed25519` and ends with your email)

Step 5: Add SSH key to GitHub

Go to **GitHub** → Settings → **SSH and GPG keys**





1. Click **New SSH key**
2. Title: `Windows Laptop`
3. Paste the key
4. Click **Add SSH key**

Step 6: Change your repo to use SSH (VERY IMPORTANT)

Go to your project folder in **Git Bash**.

Check current remote:

```
git remote -v
```

If you see `https://github.com/...`, change it:

```
git remote set-url origin git@github.com:USERNAME/REPO_NAME.git
```

Example:

```
git remote set-url origin git@github.com:padmavathynarayanan/Padmavathy-Python-Training.git
```

Step 7: Test SSH connection

```
ssh -T git@github.com
```

You should see:

```
Hi USERNAME! You've successfully authenticated...
```

SSH is working!

In case of Error in SSH verification, follow the below steps:

"Host key verification failed" is a **known SSH hiccup on Windows** - User system has an **old or corrupted SSH fingerprint** for GitHub.

Fix it (in **Git Bash**)

Step 1: Remove old GitHub SSH entry

```
ssh-keygen -R github.com
```

You should see:

```
Host github.com found: line removed
```

This clears the bad key.

Step 2: Reconnect and accept new host key

```
ssh -T git@github.com
```

When prompted:

```
Are you sure you want to continue connecting (yes/no)?
```

Type **yes** and press Enter.

Expected message:

```
Hi USERNAME! You've successfully authenticated...
```

(For **GitHub**, this is correct behavior.)

Step 8: Push code (no popups now)

```
git clone https://github.com/UserName/Repo.git
git pull
# add a file/edit/delete the existing file
git status
git add .
git status
git commit -m "commit message"
git status
git log
git push origin main
```

- No username
- No password
- No token
- No popup